PROPOFOL

High-alert drug

Strict aseptic handling

CHEMICAL NAME 2,6-diisopropylphenol (2,6-bis(propan-2-yl)phenol) in lipid emulsion
CLASS Short-acting intravenous general anaesthetic (sedative-hypnotic)
TRADE NAMES Diprivan · Propoven · Fresofol

THERAPEUTIC USES / INDICATIONS

- Induction and maintenance of general anaesthesia.
- Sedation in ICU for ventilated patients.
- Procedural sedation (endoscopy, cardioversion).

ADMINISTRATION

- **IV only** — bolus or infusion via syringe pump.
- White lipid emulsion: strict asepsis; discard vials 6 h after opening, infusion sets every 12 h.
- Pain on injection: use a large vein or lidocaine pre-treatment.
- Only by personnel trained in airway management, with resuscitation equipment.

DOSAGES

- **Induction (adult):** 1.5–2.5 mg/kg titrated (lower in elderly: 1–1.5 mg/kg).
- **Maintenance:** 4–12 mg/kg/h.
- **ICU sedation:** 0.3–4 mg/kg/h; avoid >4 mg/kg/h (propofol infusion syndrome).
- **Children (induction >1 month):** 2.5–4 mg/kg.

PHARMACOKINETICS

- **Onset:** 30–40 seconds. **Duration:** 5–10 min after single dose (rapid redistribution).
- Metabolised in liver and extrahepatic sites.
- Excreted in urine. **t½:** context-sensitive; rapid awakening even after infusion.

PHARMACODYNAMICS

- Potentiates **GABA-A** receptor → CNS depression, hypnosis.
- Reduces cerebral metabolic rate and ICP; antiemetic effect.
- No analgesic effect.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Pain on injection.
- Hypotension, bradycardia.
- Apnoea after induction; involuntary movements.

ADVERSE EFFECTS (SERIOUS)

- Profound hypotension and respiratory depression.
- Propofol infusion syndrome: metabolic acidosis, rhabdomyolysis, hyperkalaemia, heart failure (high doses >48 h).
- Hypertriglyceridaemia, pancreatitis (lipid load).
- Sepsis from contaminated emulsion; green urine (harmless).

CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity to propofol or components (soya, egg lecithin; usually safe in egg allergy).
- Sedation of children ≤16 yr in ICU.
- **Caution:** hypovolaemia, cardiac disease, elderly, lipid disorders.

KEY INTERACTIONS

- **Opioids, benzodiazepines:** additive hypotension and respiratory depression.
- Reduces dose requirement of other anaesthetics.

NURSING MANAGEMENT

- Continuous monitoring: SpO_2, ECG, BP, RR, capnography during sedation.
- Ensure airway, oxygen, suction and resuscitation equipment at bedside.
- ICU: monitor sedation score daily (sedation hold), triglycerides, CK, lactate, pH.
- Count lipid calories in nutrition plan.
- Aseptic technique; label and change lines on schedule.

PATIENT EDUCATION

- You will fall asleep within seconds and wake quickly.
- Do not drive, sign documents or drink alcohol for 24 h after sedation.
- Injection may sting briefly.

KEY POINTS

- Rapid on, rapid off IV anaesthetic; no analgesia.
- Strict asepsis — lipid emulsion supports bacterial growth.
- Causes hypotension and apnoea; airway skills needed.
- Infusion syndrome at high doses.

SOURCES CHECKED **[79]** Propofol 1% (10 mg/ml) emulsion for injection or infusion (UK SmPC) **[80]** Diprivan, IV (US label) **[81]** propofol, IV (US label) *Full references p. 432–457*